

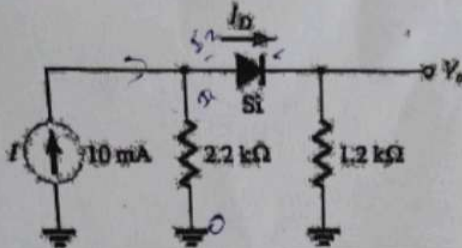
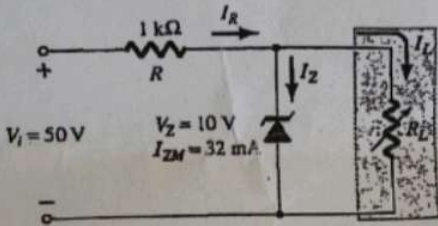
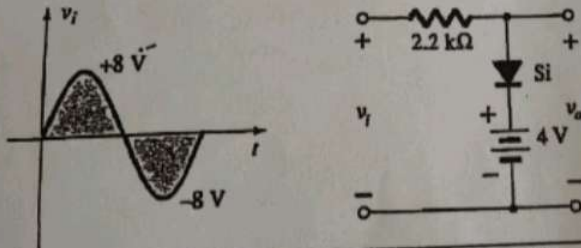
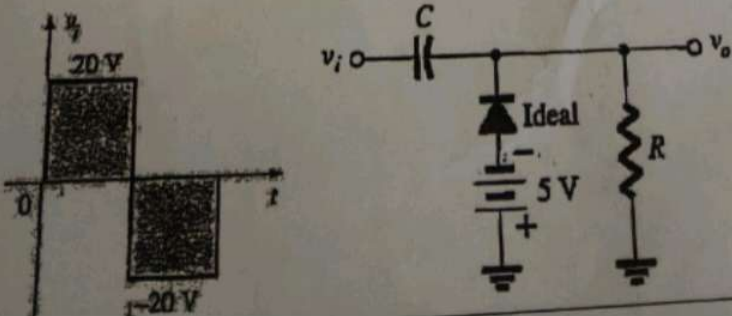


Harcourt Butler Technical University Kanpur

First
Mid Sem
Exam

Branch	BE/CH/PT/PL/OT/LT/FT/BioTech./BS-MS		Program	B.Tech
Course Name	Introduction to Electronics Engineering		Semester	
CourseCode	NET 102		Year	II
Time: 1:00Hr	Answer All Questions		MaximumMarks	15
Knowledge Level(KL)	K1:Remembering	K3:Applying	K5:Evaluating	
	K2:Understanding	K4:Analysing	K6:Creating	

Note: If any value is missing take appropriate value and solve the questions

Q.No	Questions	Marks	COs	KL
1	Draw I-V characteristics of Ideal diode. Determine V_O and I_D in below circuit 	3	CO1	K1
2	For below Zener diode circuit, find the range of R_L and I_L. 	3	CO1	K4,5
3	Draw the output (v_o) waveform for given circuit- 	3	CO1	K3,5
4	Define Peak Inverse Voltage? For bridge rectifier value of the parameters (i) form factor, (ii) ripple factor, (iii) peak value & (iv) transformer utilization Factor.	3	CO1	K1,5
5	Draw the output (v_o) wave form for below figure- 	3	CO1	K4,5

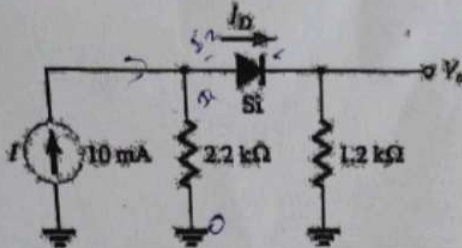
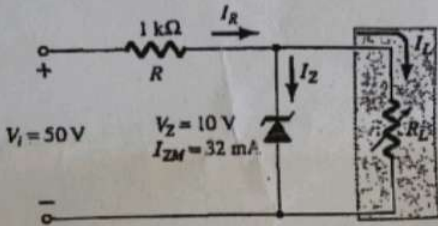
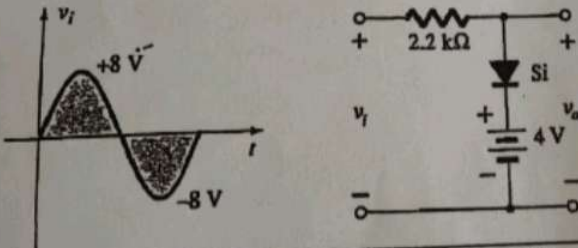
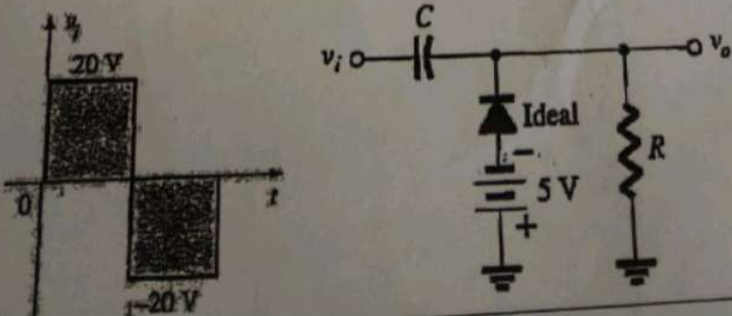


Harcourt Butler Technical University Kanpur

First
Mid Sem
Exam

Branch	BE/CH/PT/PL/OT/LT/FT/BioTech./BS-MS		Program	B.Tech
Course Name	Introduction to Electronics Engineering		Semester	
CourseCode	NET 102		Year	II
Time: 1:00Hr	Answer All Questions		MaximumMarks	2025-26
Knowledge Level(KL)	K1:Remembering	K3:Applying	15	
	K2:Understanding	K4:Analysing	K5:Evaluating	
			K6:Creating	
Note: If any value is missing, take appropriate value.				

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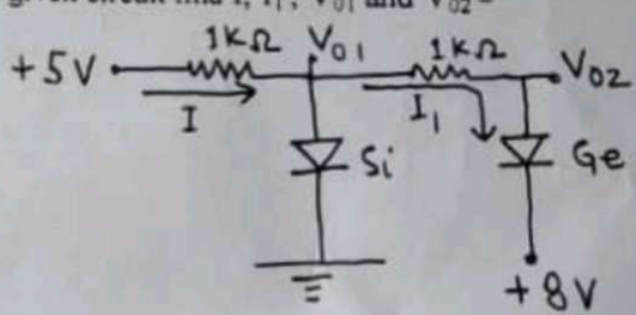
Q.No	Questions	Marks	COs	KL
1	Draw I-V characteristics of Ideal diode. Determine V_O and I_D in below circuit 	3	CO1	K1
2	For below Zener diode circuit, find the range of R_L and I_L . 	3	CO1	K4,5
3	Draw the output (v_o) waveform for given circuit- 	3	CO1	K3,5
4	Define Peak Inverse Voltage? For bridge rectifier value of the parameters (i) form factor, (ii) ripple factor, (iii) peak value & (iv) transformer utilization Factor.	3	CO1	K1,5
5	Draw the output (v_o) wave form for below figure- 	3	CO1	K4,5

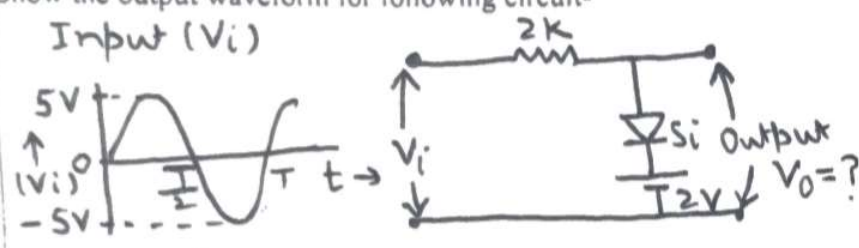
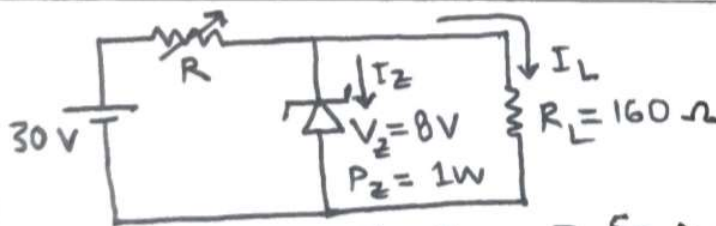


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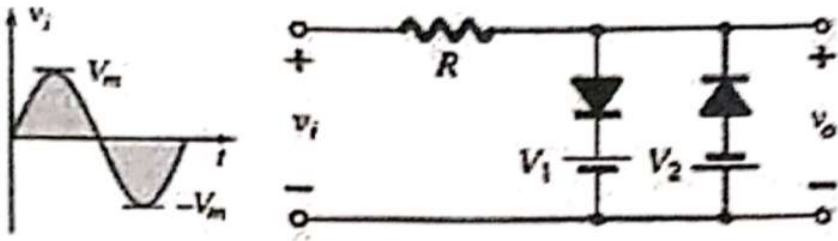
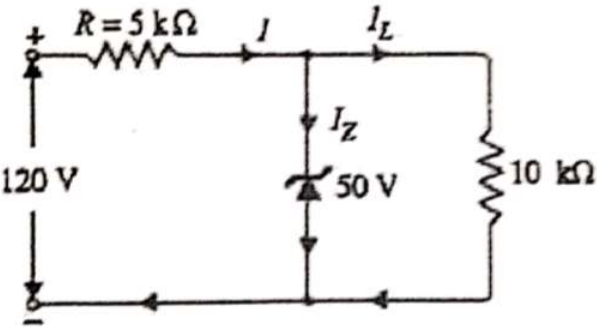
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TERM-I

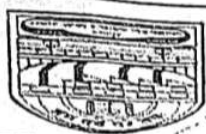
Branch	CE, ME, EE, ET, CS, IT		Program	B.Tech
Course Name	Introduction to Electronics Engineering		Semester	1
Course Code	NET 101		Year	1 (2025-26)
Time: 1:00Hr	Answer All Questions		Maximum Marks	15
Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating	
	K2: Understanding	K4: Analysing	K6: Creating	

Q.No	Questions	Marks	COs	KL
1.(a)	Explain the comparison of the conductivities of Insulator, Semiconductor and Metal on the basis of their energy band diagrams with appropriate diagram.	3	1	K2
(b)	The forward current through a Si diode is 10mA at room temperature. The corresponding forward voltage is 0.75 V. Find the value of reverse saturation current.	2	1	K3
2.(a)	Explain the reverse and forward bias operation of PN junction diode in brief, also draw the V-I characteristic curve of diode.	4	1	K1
(b)	Show that Ideal diode behaves like a switch, also draw it's characteristic	2	1	K4
3.	For given circuit find I , I_1 , V_{01} and V_{02} - 	4	1	K5

		Harcourt Butler Technical University Kanpur		MID TERM-II
Branch	Electronics Engineering	Program	B.Tech	
And Course Name	Introduction to Electronics Engineering	Semester	I	
Course Code	NET 101	Year	I (2025-26)	
Time: 1:00Hr	Answer All Questions	Maximum Marks	15	
Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating	
	K2: Understanding	K4: Analysing	K6: Creating	
Q.No	Questions	Marks	COs	KL
1.	Explain the mechanisms to explain the breakdown mechanisms in diode characteristic.	4	1	K2
2.	Establish relation between α and β . Evaluate change in α , if β is increased from 51 to 80. Or Show the output waveform for following circuit- 	3	1,2	K3
3.	Draw and explain input and output characteristic curves for the Common Base Configuration. Or Explain the working of Bridge Rectifier. Also evaluate efficiency and the PIV.	4	1,2	K2
4.	 For given circuit, $I_{Z(min)} = 5mA$, $P_{Z(max)} = 1Watt$, Find the range of R , such that load voltage is 8V constant.	4	3	K5

Course Outcomes	CO1	To understand the basic concept of diodes, and use the diode as a circuit element for different applications.
	CO2	To understand the working of BJT, FET, OP-amp and their application.
	CO3	To design the simple digital circuits using different logic gates.
	CO4	To identify the errors while making electronic measurements and to understand the working of different types of transducers.
	CO5	To understand the working principle of electronic instruments and displaying it on electronic devices.

Q. NO	Questions	Marks	CO
1.(a)	Explain the working of center tapped full wave rectifier. Also calculate PIV .	3.5	1
(b)	For the following circuit assuming both diodes are ideal -  Find the output waveform V_o, if $V_m = 9\text{v}$, $V_1 = 5\text{v}$ and $V_2 = 6\text{v}$	3.5	1
(c)	For the given circuit find (i) Current through series resistance R (ii) Find zener current I_Z (iii) Find load current I_L . 	3	1



Harcourt Butler Technical University Kanpur

FIRS
SEM
202

Branch	CE/CSE/ET/IT/ME	Program	B.
Course Name	Introduction to Electronics Engineering	Semester	
Course Code	NET-101	Year	
Time: 1:00 Hr	Answer All Questions	Maximum Marks	
Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating
	K2: Understanding	K4: Analysing	K6: Creating

Note: Assume any data if necessary.

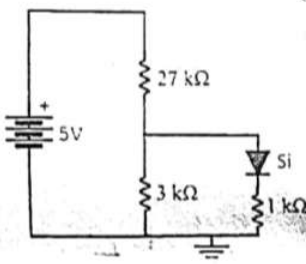
Q. No	Questions	Marks	COs
1	Explain the working principle of a P-N junction diode under open circuit condition. Also discuss how the depletion region is formed.	3	CO1
2	Explain the working of a P-N junction diode under forward bias and reverse bias conditions along with the V-I characteristics.	3	CO1
3	Calculate the voltage (V_D) across the diode and the current (I_D) through the diode of Figure Q. 3. 	3	CO1
4	Differentiate between Avalanche breakdown and Zener breakdown in a P-N junction diode. Explain the mechanisms behind each type of breakdown and the conditions under which they occur.	3	CO1
5	Explain the operation of center-tapped full wave rectifier with the help of a circuit diagram. Define its PIV.	3	CO1

Figure Q. 3

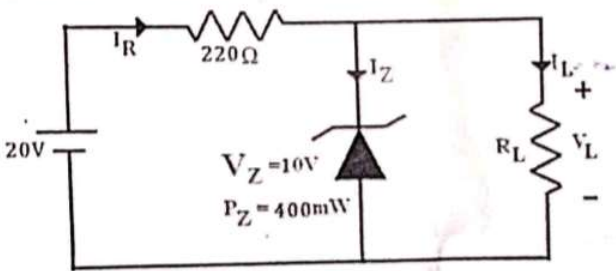


Harcourt Butler Technical University Kanpur

End Sem
Exam
2024-25

Branch	CE/CSE/EE/ET/IT/ME		Program	B. Tech
Course Name	Introduction To Electronics Engineering		Semester	I
Course Code	NET-101		Year	1 st Year
Time: 2:30 Hr	Answer All Questions		Maximum Marks	50
Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating	
	K2: Understanding	K4: Analyzing	K6: Creating	

Note: Assume any data if necessary. Attempt all parts of question at one place.

Q. No	Questions	Marks	COs	KL
1(a)	A 230 V, 50 Hz A.C. is applied to the primary of a 5:1 step-down transformer in a bridge rectifier having a load resistor of $500\ \Omega$ assuming diodes are ideal diode. Find the following: (i) D. C. output voltage (ii) D. C. power at load (iii) PIV of each diode (iv) Output frequency	5	CO1	K1, K2
1(b)	Distinguish between Avalanche and Zener breakdown mechanism. Calculate I_R , I_Z , I_L and V_L for the network given below if $R_L=470\ \Omega$. 	5	CO1	K4, K5